Alchemy

Documentation Infrastructure Audit Report

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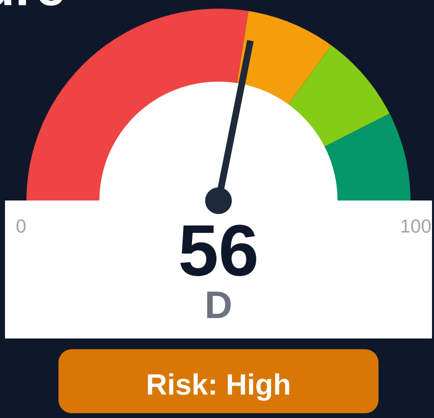


Key Metrics

Pages scanned: 7661

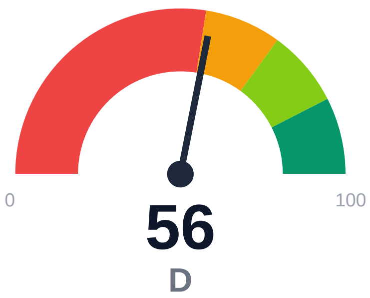
Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 11
- API usage-doc coverage is low: 16.67% (6415 API pages detected)
- Freshness visibility is weak: only 14.87% pages show last updated

Executive Summary



Alchemy's documentation spans 7,661 crawled pages across a multi-project topology with 83.21% crawl coverage of the 10,215 discovered pages. API reference coverage is critically low at 16.67% despite 6,415 API pages detected, indicating significant gaps in usage documentation. Only 14.87% of pages display last-updated metadata, creating trust and staleness risk across 75.59% of content. Example code reliability is estimated at 70.19%, and overall retrieval readiness scores 69.77 (moderate band) with answerability at just 51.39%. The site has 11 confirmed broken internal links, all within user-facing docs.

External posture: SEO/GEO issue rate **4.9%**, public API coverage **16.7%**, broken links **11**.

56 Audit Score	**F** Grade	**High** Risk Band
7661 Pages Scanned	**11** Broken Links	**4.9%** SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://alchemy.com/docs	7661	11	16.7	70.2	4.9

Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	**92+**	**-36**
Industry average (developer docs)	**85**	**-29**
Minimum acceptable	**70**	**-14**
Your score	**56**	**-**

Gap to industry average: **29 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

API Coverage (Public)

16.7%

SEO/GEO Issue Rate (lower is better)

4.9%

Example Reliability (estimate)

70.2%

Freshness Visibility (last-updated metadata)

14.9%

Internal Metrics (from scorecard)

API Coverage (Internal)

16.7%

Example Reliability (Internal)

70.2%

Docs-Contract Drift

27.1%

Pipeline Opportunity Fit (Deterministic)

This section maps cold-audit signals to problem classes that the pipeline can remediate directly. No LLM inference is used in this mapping.

LLM Retrieval Readiness (deterministic proxy)

69.8%

Audit Confidence (crawl scope + verification quality)

70.0%

Readiness band: **Moderate**. Confidence: **Medium**. Open remediation opportunities: **4**.

SEO/GEO Quality Signal (inverse issue rate)

80.6%

Example Reliability Signal

70.2%

Freshness Visibility Signal

14.9%

API Coverage Signal

16.7%

LLM retrieval readiness breakdown (deterministic)

Chunkability

84.1%

Answerability

51.4%

Citationability

99.9%

Stale Risk (inverse)

24.4%

Evidence coverage: **72.9%** (11928/16371 key claims backed by deterministic evidence signals).

Highest-value remediation targets

Problem	Severity	Pipeline solution path
Internal documentation link integrity issues	HIGH	Weekly link governance via automated audits and consolidated remediation queue
Code example reliability risk	MEDIUM	Snippet lint + smoke checks + protocol self-verify coverage
Weak freshness and lifecycle visibility	MEDIUM	Lifecycle management, stale detection, and weekly governance cadence
API reference coverage and drift risk	HIGH	API-first contract flow, validators, regression gates, and drift checks

Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
$355,626 annual exposure	**$449,495** annual exposure	**$606,863** annual exposure
Remediation: **$1,805**	Remediation: **$4,702**	Remediation: **$7,600**
Monthly loss: **$3,562**	Monthly loss: **$11,144**	Monthly loss: **$24,016**
Opportunity: **$25,923**/mo	Opportunity: **$25,923**/mo	Opportunity: **$25,923**/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-API-COVERAGE	Undocumented API operations	HIGH	$2,508	$950
F-FRESHNESS	Documentation freshness debt	HIGH	$969	$475
F-DRIFT	Code/docs contract drift	HIGH	$1,642	$570
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	MEDIUM	$1,938	$808
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	$1,949	$855
F-LAYERS	Missing required doc layers	LOW	$1,539	$712
F-TERMINOLOGY	Terminology inconsistency	LOW	$598	$332

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-API-COVERAGE	Undocumented API operations	HIGH
F-FRESHNESS	Documentation freshness debt	HIGH
F-DRIFT	Code/docs contract drift	HIGH
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM

Recommended Actions

1. Generate and backfill usage examples for the 5,348 API pages lacking coverage (83.33% gap) starting with top 200 most-visited endpoints [impact: critical, effort: high]

2. Add last-updated timestamps to all 6,522 pages missing freshness metadata (85.13%) via automated git commit date injection [impact: high, effort: low]

3. Restructure the 3,735 pages with low answerability (48.61%) to include explicit Q&A; sections, summaries, or structured headings [impact: high, effort: medium]

4. Audit and fix all 11 broken internal links, then implement automated link checking in CI/CD to prevent recurrence [impact: medium, effort: low]

5. Implement automated testing for code examples across the 2,285 pages with reliability concerns (29.81% failure estimate) [impact: high, effort: high]

Expert Analysis: Strengths and Risks

Strengths

+ High crawl coverage at 83.21% with 7,661 pages successfully indexed from 10,215 discovered
+ Strong citationability at 99.93% enables reliable source attribution for AI and search
+ Chunkability at 84.06% supports effective content segmentation for retrieval systems
+ Evidence coverage at 72.86% with 11,928 of 16,371 claims backed by supporting material
+ 100% actionability coverage ensures content provides clear next steps for users

Risks

- API reference coverage at 16.67% leaves 5,348+ API pages without adequate usage documentation, blocking developer onboarding
- Answerability at 51.39% means nearly half of pages lack structure for direct question responses, degrading AI assistant and search quality
- Freshness metadata missing on 85.13% of pages (6,522 pages) creates user trust issues and prevents stale content identification
- Stale risk flagged on 75.59% of content (5,791 pages) without visible update signals, risking outdated guidance in production use
- Example reliability at 70.19% suggests ~30% of code samples may be broken, outdated, or untested, eroding developer trust
- 11 broken internal links disrupt navigation and signal maintenance gaps to users and search engines
- SEO/geo issues affect 4.86% of pages, potentially limiting discoverability in key markets or search contexts

Limitations

* External audit cannot verify runtime correctness of the 70.19% example reliability estimate; actual code execution, dependency versions, and API behavior require live testing infrastructure.
* Crawl coverage at 83.21% means 1,554 discovered pages were not examined; access restrictions, JavaScript rendering issues, or crawler traps may hide additional quality issues.
* API coverage detection at 85% confidence may misclassify pages; the 16.67% reference coverage figure depends on heuristic detection of usage vs. reference content and may undercount embedded examples.
* Stale risk at 75.59% is inferred from missing freshness signals, not content accuracy; pages without timestamps may be current, and timestamped pages may contain outdated technical details.
* Answerability and chunkability scores reflect structural patterns, not semantic correctness; pages may score well but contain incorrect, incomplete, or misleading information undetectable without domain expertise.

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	95.0
support_hourly_usd	65.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	18.0
monthly_support_tickets	850.0
docs_related_ticket_share	0.28
avg_ticket_handling_minutes	22.0
monthly_doc_influenced_evals	180.0
eval_to_customer_rate	0.12
avg_customer_monthly_value_usd	2400.0
docs_friction_revenue_sensitivity	0.35

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH

F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	LOW

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.